Q1) Answer the following MCQs or Short Questions:

a) Pre-order traversal of a binary tree gives A B D E C F and in-order traversal gives D B E A F C. What is the right child of the root? [1]
i) B ii) F iii) C iv) D

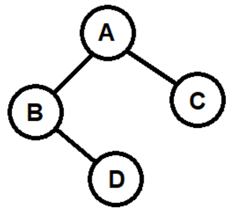
b) A binary tree has 12 nodes. What is the minimum possible height of this tree [1]
i) 2 ii) 3 iii) 4 iv) 11

c) Two binary trees have the same in-order traversal sequence. Does this mean they are structurally the same? Justify with a counter example by drawing an example tree. [2]

Ans:

d) The binary tree below has height h = 2. Show the array representation of this tree (1-indexed), clearly indicating which slots are empty (null). [2]

Ans:



Q2) Write a function called **countBalancedSum(root)** that takes the root of a binary tree and returns the number of **balanced-sum** nodes. A node has **balanced-sum** if: sum of values in its left subtree == sum of values in its right subtree. The leaf nodes are not considered as balanced-sum since they don't have any subtrees. [12]

Note: Assume BTNode class is defined with three instance variables: elem(int), right(BTNode), left(BTNode). You must use recursion. You are not allowed to create any other data structure or use static or instance variables. You are allowed to use helper functions if needed.

Given Tree	Output	Explanation
10 / \\ / \\ 5 7 / \\ / \\ 12 8 9 6 \\ 3	2	10 → left subtree sum: 25, right subtree sum: 25 → **10 has balanced-sum** 5 → left subtree sum: 12, right subtree sum: 8 → 5 has NO balanced-sum 7 → left subtree sum: 9, right subtree sum: 9 → **7 has balanced-sum** 6 → left subtree sum: 0, right subtree sum: 3 → 6 has NO balanced-sum 12,8,9,3 were ignored since they were leaf nodes **Total balanced-sum nodes: 2**

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                 Java                     |                Python
static int countBalancedSum( BTNode root ){    |     def countBalancedSum( root ):
```